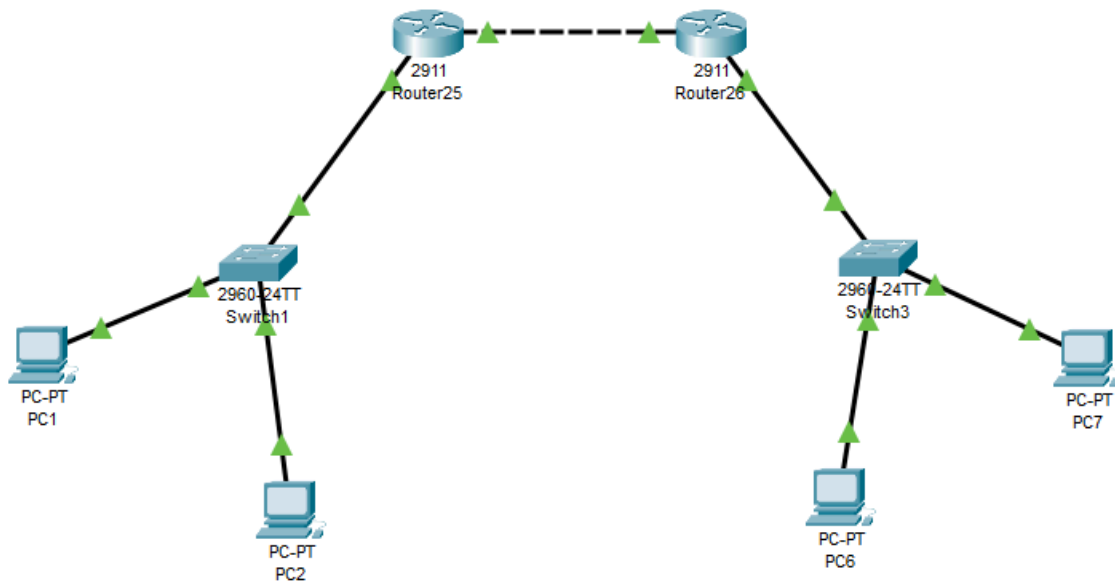


Mini ISP WITH VLAN

This is the main project structure.

PC1, PC2 -> Switch1 -> Router25 <-> Router26 <- Switch3 <- PC6, PC7



Pic: Mini ISP with Vlan

1. Initial Setup

All end devices (PCs) were first configured with appropriate IPv4 addresses and default gateways according to their respective network segments. After completing host configuration, basic connectivity between networks was verified using ICMP ping tests.

2. Initial Router Configuration (Inter-Network Routing)

Router25 Configuration

Router25 was configured to support two interfaces: one connected to the local LAN (192.168.10.0/24) and the other connected to the inter-router link (10.0.0.0/30).

enable

configure terminal

interface g0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

interface g0/1

ip address 10.0.0.1 255.255.255.252

no shutdown

ip route 192.168.20.0 255.255.255.0 10.0.0.2

Router26 Configuration

Router26 was configured similarly for its local LAN and inter-router connection.

enable

configure terminal

interface g0/0

ip address 192.168.20.1 255.255.255.0

no shutdown

interface g0/1

ip address 10.0.0.2 255.255.255.252

no shutdown

ip route 192.168.10.0 255.255.255.0 10.0.0.1

3. Initial Connectivity Testing

After completing router configuration, end-to-end connectivity was verified. Ping tests between PCs in different networks (PC2, PC6, and PC7) were successful. Similarly, reverse connectivity tests from PC6 and PC7 to PC1 were also successful, confirming proper static routing between the two networks.

4. VLAN Configuration on Switch1

To segment the LAN into different broadcast domains, VLANs were configured on Switch1.

Switch1 VLAN Setup

```
enable  
configure terminal
```

```
vlan 10  
name HR
```

```
vlan 20  
name IT
```

Port Assignment

```
interface fa0/24  
switchport mode access  
switchport access vlan 10
```

```
interface fa0/23  
switchport mode access  
switchport access vlan 20
```

Trunk Configuration

```
interface g0/0  
switchport mode trunk
```

5. Router-on-a-Stick Configuration (Inter-VLAN Routing)

To enable communication between VLAN 10 and VLAN 20, router-on-a-stick configuration was implemented on Router25.

Router25 Subinterfaces

enable

configure terminal

interface g0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

interface g0/0.20

encapsulation dot1Q 20

ip address 192.168.30.1 255.255.255.0

interface g0/0

no shutdown

6. Additional Routing Configuration on Router26

To ensure reachability to the newly introduced VLAN 20 subnet, a static route was added on Router26.

```
ip route 192.168.30.0 255.255.255.0 10.0.0.1
```

7. Final Verification

After implementing VLANs and inter-VLAN routing, connectivity tests were performed again. Successful ping responses confirmed:

- Inter-VLAN communication between VLAN 10 and VLAN 20
- Continued connectivity between Router25 and Router26 networks
- Full end-to-end communication across all configured subnets

8. Conclusion

The network was successfully configured using a combination of static routing and inter-VLAN routing. VLAN segmentation improved network organization and security, while trunking and subinterfaces enabled efficient communication between different VLANs and routed networks.

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